

for 2-3 years of product demand in the design phase, helps to address this issue. Having a strong vendor relationship for component availability and shortage information is a must-do in the current scenario.

### **Overcoming lab access limitations and remote work challenges**

Getting 100% lab access can be a challenge due to lockdown, travel restrictions, or engineers working from home options. It enables engineers to spend time and automate lab instruments for all test cases. Engineers can remotely access the instruments while working from home and still perform all planned test cases. This avoids human errors due to manual testing and helps to improve overall test convergence and test time. It helps engineers to store test results using automation for future builds.

### **Reusable schematic and layout blocks**

Designing hardware schematics and layouts in the form of reusable blocks that can be used for all future designs will help hardware designers to build error-free designs and to use the same components in all future designs. It enables designing at a low cost, as the same components require huge quantity for production builds. Hardware blocks should be error-free and fully validated/characterised before they are certified as reusable blocks. This method will help to standardise components and other design accessories across all products.

### **Component part database**

80% of hardware design and production DFM/DFA issues are due to using incorrect/unverified footprints in design. Building a component part database with all certified and proven footprints will help avoid

DFM/DFA issues in production builds. These footprints should be linked with the production build environment through PLM (Product Life Cycle Management) system so that they will be reusable across all designs. These PLM tools can be linked/interconnected with component life cycle checking tools so that engineers can track Active/EOL status using the PLM system, and it will help to identify EOL parts in the early phase of design.

### **Hardware tool automation**

Hardware engineers can use tickle scripts to automate schematic and layout tools and implement different design rules checking (DRC). It will help to reduce manual verification-related issues. This automation will reduce design as well as verification cycles and avoid schematic and layout-related issues.

### **Designing quick prototyping using development kits, conductive ink, and 3D print**

Building quick prototyping using development kits/evaluation kits and 3D print (in case of mechanical design) will help the hardware designer prove concepts before they go into the final design. Designing a quick prototype using conductive ink for a small design with fewer components avoids fabrication and PCB lead time costs.

### **ECAD (electronic tool) and MCAD (mechanical tool) collaboration**

Creating 3D step files for every footprint used in the design and integrating them with the mechanical tool will help designers to review complete product design in 3D form and do mechanical integrity checks with other interconnecting boards. It avoids connector mating as well as accessibility-related issues. It helps designers to review boards in end

product form factors with reviewers so that they will get exactly the same look and feel.

### **Driving standardisation through design templates**

Engineers can make different templates and checklists for hardware design files like schematic templates, BOM templates, FR4 layout fabrication templates, flex layout fabrication templates, and mechanical design templates to reduce design errors. It helps engineers to implement best practices across all designs. By designing a proper layout template engineers can avoid maximum CAM queries and fabrication issues. With the help of proper design templates, engineers can communicate all design challenges and constraints to the fabrication/manufacturing house without phone calls. It also helps engineers to produce designs globally at multiple locations. Using proper templates, hardware designers can make capability matrices for assembly and fabrication vendors, which will help designers to select the best vendor for each job based on design complexities.

### **Pre- and post-circuit simulation**

Pre- and post-circuit simulation tools will help the hardware designer to verify low-speed as well as high-speed interface signals using simulation tools. This avoids multiple spins of boards as well as signal integrity issues. It's a challenge for hardware engineers to select the correct and accurate simulation tool to get the best results.

So, there you have it—a toolbox brimming with insights to revolutionise your hardware design journey. As we step into the future of hardware design, remember: it's not just about circuits; it's about adaptability, innovation, and turning challenges into triumphs. **EFY**